

pixel recording an image and transmitting this to the patient via the optic nerve. Such technology is not restricted to vision and is applicable to many other types of sensation.

A microarray thin film can also be integrated into bed linen and patient dormitories, and can operate in similar ways to monitor and identify abnormalities in body temperature, as

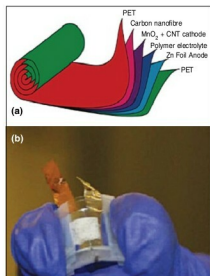


Fig. 12: (a) battery structure, (b) optical micrograph of a flexible battery during flexing

well as sweat-elemental analysis. As with the lab-on-a-chip (LOC) applications, these thin films will become a key component of our approach to next-generation healthcare. Heat distribution in the body, sweat content and frequency or postural pressure on part of the body can all reveal vital information on pathological symptoms or recovering stages.

Further applications of microarray systems based on such flexible thin-film technology are a facilitator for artificial noses and tongues, as shown in Fig. 11. Sensory receptors in olfactory (nose) and gustatory (taste) systems have a range of chemical receptors. Many of these receptors sense particular chemical properties, including acidity, salt concentration and enzyme affinity.

It can also be used to monitor the pulses from brain and analyse the changes and design the treatment based on the present situation of the patient.

In automotive industry

The combustion reaction has fueled the development of humanity since the discovery of fire. The main reasons for the ubiquity of combustion are twofold. The first relates to its intrinsic mass and volume energy densities. For example, one kg of crude oil contains nearly 50MJ of chemical potential energy. The second relates to storage and portability. Crude oil happens to be liquid at surface conditions, making it easy to store, transport and convert, which is fundamental for applications such as transportation.

Although electric vehicles trace their invention to the mid-1800s, their widespread adoption has not yet materialised. This is partly due to the high energy

density (J/kg) of fossil fuel compared to the batteries, with the implication that electric vehicles require unfeasibly heavy and large batteries to store equivalent energy as a tank of fuel. Besides low energy density, batteries suffer from low power density (W/kg), leading to electric vehicles with comparatively low performance when compared to equivalent fuel-powered vehicles. Despite these issues, increasingly scarce fossil fuel resources, as well as environmental incentives, have increased the adoption of electric vehicles, fueling research and development in this field.

Advances in thin-film battery technology through the use of nanostructures for enhanced energy density and hybrid supercapacitor allowed increased energy and power densities. Lightweight substrates, such as polyethylene terephthalate (PET) and paper, have led to a reduction in battery weight. Flexible, thin-film technology is especially beneficial in this instance, as it allows batteries to be moulded into suitable shapes at relatively low costs.

Flexible thin-film technology may also find applications in road signs and markings. Intelligent roads will be engineered with the aim of improving road safety, lowering road congestion and energy consumption. The road and vehicle will be able to interact to dynamically adjust either party to energetically optimise their systems. The advantage of flexible thin-film technology is its mechanical durability and ease of inexpensive integration within the existing road networks.

In energy management and mobile devices

The innovative gadgets that we are presently experiencing are made possible by flex circuits in the industry. Today, mobile and other electronic gadgets manufacturers are designing the products keeping in view the future scope of the product and its use in more than one way. The ease of handling the device is also an important factor that needs to be noted while



Fig. 13: The Morph mobile phone concept, where flexibility and transparency become important (Courtesy: Nokia Research)

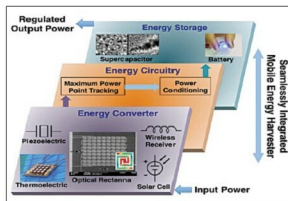


Fig. 14: Components of a flexible, wearable mobile energy-harvesting system